Abhinav Mahajan

abhinavm@andrew.cmu.edu | (412) 608-7868 | abhinav-mahajan10.github.io | linkedin.com/in/abhinav-mahajan10

EDUCATION

Carnegie Mellon University — **Pittsburgh, PA**
Master of Science in Computer Vision | GPA: 4.33/4.0 — *December 2026*

International Institute of Information Technology (IIIT Bangalore) — **Bangalore, India**
(Dual Degree) Master and Bachelor of Technology in Electronics and Communication Engineering — *July 2025*
GPA: 3.63/4.00; Dean's Merit List; Teaching Assistant for courses Machine Learning (AI 511), Visual Recognition (AI 825)

SKILLS

Programming Languages: Advanced: Python; Basic: C++, Java, C
Libraries: Advanced**:** PyTorch, NumPy, Pandas, Diffusers, OpenCV, Transformers, trl; Basic: Tensorflow
Specialized: Diffusion Models, LLMs. Multimodal Models, Neural Radiance Fields, Gaussian Splatting, AWS, Docker

PROFESSIONAL EXPERIENCE

Amazon — **Bangalore, India**
Applied Scientist Intern [Advisor: Arindam Sarkar, Dr. Prakash Mandayam Comar] — *January 2025 - June 2025*

- Achieved 3x recall in next-purchase categories prediction by developing LLM based customer interest scoring pipeline by a novel technique for mitigating exposure bias and enabling extremely-long sequential context compression.
- Augmented a dual-encoder propensity scoring system, extracting prototypical classification embeddings from the generative model to enable bi-directional ranking of customers and categories for personalized ad targeting at scale.
- Paper under review at KDD 2026; Received full-time offer as Applied Scientist-1.

Adobe Research — **Bangalore, India**
Research Intern [Advisor: Dr. KJ Joseph, Dr. Balaji Vasan Srinivasan] — *May 2024 - July 2024*

- Developed Components-to-Design framework for harmonizing multimodal design assets (images, text, shapes) into aesthetic graphic designs (digital Posters/Flyers) using Large Multimodal Models and diffusion models.
- Designed a training-free identity-preserving composition method through cross-attention and self-attention matrix manipulation for LMM's, significantly improving foreground identity retention while maintaining semantic coherence.
- Paper in preparation for ICML 2026; Patent in provisional stage; Received full-time offer as MLE-1 in Firefly team.

Adobe Research — **Bangalore, India**
Research Intern [Advisor: Dr. KJ Joseph, Dr. Balaji Vasan Srinivasan] — *May 2023 - August 2023*

- Trained a self-supervised graphic design aesthetics scoring model using Siamese networks and contrastive learning to eliminate human annotation bias, achieving 94.97% accuracy in ranking graphic designs.
- Developed a novel genetic algorithm for layout refinement optimizing on the aesthetics scorer, outperforming all baselines; System deployed in Adobe Express; Paper accepted at WACV 2025 (Round 1, 13% acceptance rate).

PUBLICATIONS

[1] **Abhinav Mahajan*** et al. *Generative XMC for Customer-Interest Category Propensity*, **Under Review**
[2] **Abhinav Mahajan*,** Abhikhya Tripathi* et al. *Make-It-Pretty: Generating Designs from its Components,* **Under Review**
[3] **Abhinav Mahajan*,** S Goyal* et al. *Design-o-meter: Towards Evaluating and Refining Graphic Designs,* **WACV 2025**
[4] Ali Abedi*, Tracey JF Colella*, **Abhinav Mahajan** et al. *AVA: AI-driven Virtual Rehabilitation Assistant,* **WCISVR 2023**

PROJECTS

Hierarchical Activity Analysis of Users — *August 2024 - October 2024*

- Awarded the MITACS Fellowship and interned at Ontario Tech University, Canada. Developed a platform to crowd source a dataset, contributed to research on screenshot parsing with Vision Language Models and Active-Learning.

Pothole Detection and Severity Estimation — *December 2022 - January 2023*

- Won the SDAIA Smartathon Hackathon among 1,000+ submissions by training a Mask R-CNN model achieving 82% segmentation accuracy on video data and a 3D reconstruction pipeline of potholes for severity and risk assessment.

Stroke Rehabilitation Agent — *May 2022 - December 2022*

- Collaborated with University of Toronto to develop a video-to-animation pipeline enabling clinician-prescribed exercises to render as jitter-free avatar animations via novel cyclic interpolation; Paper accepted at WCISVR 2023.